1. A speaker that vibrates makes a sound wave in the air with a wavelength of 34 m. If the speed of sound waves is 340 m/s, then what is the frequency of the sound?

⊕ Known :

$\lambda = 34m$

$v = 340 m/s$

⊕ Wanted :

$f = ?$

⊕ Answer :

$v = \lambda . f$

$\frac{340}{34} = \boxed{34} f$

$f = 10 Hz$

2. A source of vibration produces 600 vibrations in 1 minute. Calculate the wavelength produced if the speed of waves in the air is 340 m/s?

⊕ Known :

$n = 600$

$t = 1 min$

$= 60 s$

$v = 340 m/s$

⊕ Wanted

$\lambda = ?$

⊕ Answer

→ $v = \lambda . f$ → find this first.

→ $f = \frac{n}{t} = \frac{600}{60} = 10 Hz$

→ $v = \lambda . f$

$\frac{340}{10} = \lambda . \boxed{10}$

$\lambda = \frac{340}{10} = 34 m$

3. What is the period and speed of a wave for a wave that has frequency 2 kHz and wavelength 17 cm?

⊕ Known

$f = 2 kHz = 2000 Hz$

$\lambda = 17 cm = \frac{17}{100} m$

⊕ Wanted :

$T = ?$

$v = ?$

⊕ Answer:

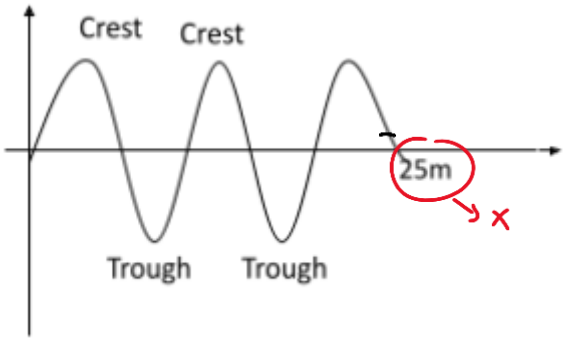
→ $T = \frac{1}{f} = \frac{1}{2000} s$

→ $v = \lambda . f$

$= \frac{17}{\cancel{100}} . 20\cancel{00}$

$v = 340 m/s$

4. The picture shows a wave that travel at 80 m/s
   a. What is the wavelength of the wave? → $\lambda = ?$
   b. What is the frequency of the wave? → $f = ?$



• Known:

$v = 80\,m/s$ $\quad n = 2.5$

$x = 25\,m$

• Wanted:

$\lambda = ?$

$f = ?$

• Answer:

→ $n.\lambda = x$

$\boxed{2.5}.\lambda = \frac{25}{2.5}$

$\lambda = 10\,m$

→ $v = \lambda . f$

$80 = 10 . f$

$f = 8\,m/s$

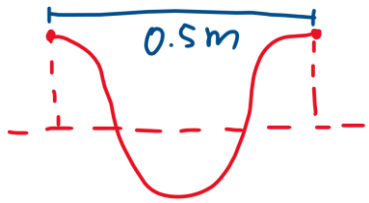
5. A tsunami wave travels in the ocean, from the measurement, it is able to travel 350 m in 5 s. What is the frequency and period of the wave if the distance between two tops of waves is 0.5 m?

• Known:

$x = 350\,m$ → considered as distance travelled by the wave

$t = 5\,s$

$\lambda = 0.5\,m$ → distance between 2 tops



• Wanted:

$f = ?$ $\quad T = ?$

• Answer:

$v = \lambda . f$

find this first

• $v = \frac{x}{t} = \frac{350}{5} = 70\,m/s$

⇒ $v = \lambda . f$

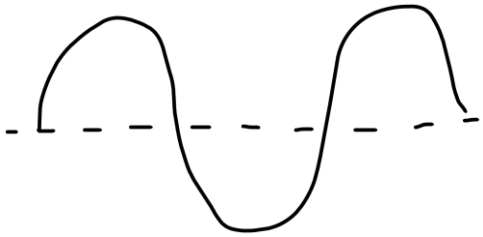
$\frac{70}{0.5} = \boxed{0.5} . f$

$f = 140\,Hz$

⇒ $T = \frac{1}{f} = \frac{1}{140}\,s$

6. 2 crests and 1 trough of waves are produced in 1.5 s.
   a. What is the frequency of the wave?
   b. What is the period of the wave?

• Known :



2 crests and 1 trough ⇒ $n = 1.5$

$t = 1.5\ s$

• Wanted :

$f = ?$ $T = ?$

• Answer

$f = \frac{n}{t} = \frac{1.5}{1.5} = 1\ Hz$

$T = \frac{1}{f} = 1\ s$

7. A piece of cork floats above sea water. The cork moves upside down 100 times in 40 s. In the same time the sea wave has traveled 60 m. What is the wavelength of the wave?

• Known :

$n = 100$ → 1 upside down means 1 vibration / wave

$t = 40\ s$

$x = 60\ m$

• Wanted :

$\lambda = ?$

• Answer :

$n . \lambda = x$

$100 . \lambda = 60$

$\lambda = \frac{60}{100} = 0.6\ m$